

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education
Advanced Subsidiary Level and Advanced Level

HIGHER MATHEMATICS
MATHEMATICS

8719/03
9709/03

Paper 3 Pure Mathematics 3 **(P3)**

October/November 2004

1 hour 45 minutes

Additional materials: Answer Booklet/Paper
Graph paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is 75.
Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.
The use of an electronic calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

This document consists of 4 printed pages.



- 1 Expand $\frac{1}{(2+x)^3}$ in ascending powers of x , up to and including the term in x^2 , simplifying the coefficients. [4]

- 2 Solve the equation

$$\ln(1+x) = 1 + \ln x,$$

giving your answer correct to 2 significant figures. [4]

- 3 The polynomial $2x^3 + ax^2 - 4$ is denoted by $p(x)$. It is given that $(x-2)$ is a factor of $p(x)$.

- (i) Find the value of a . [2]

When a has this value,

- (ii) factorise $p(x)$, [2]

- (iii) solve the inequality $p(x) > 0$, justifying your answer. [2]

- 4 (i) Show that the equation

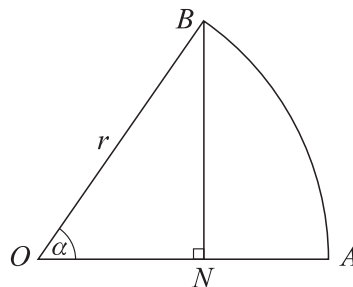
$$\tan(45^\circ + x) = 2 \tan(45^\circ - x)$$

can be written in the form

$$\tan^2 x - 6 \tan x + 1 = 0. [4]$$

- (ii) Hence solve the equation $\tan(45^\circ + x) = 2 \tan(45^\circ - x)$, for $0^\circ < x < 90^\circ$. [3]

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The diagram shows a sector OAB of a circle with centre O and radius r . The angle AOB is α radians, where $0 < \alpha < \frac{1}{2}\pi$. The point N on OA is such that BN is perpendicular to OA . The area of the triangle ONB is half the area of the sector OAB .

- (i) Show that α satisfies the equation $\sin 2x = x$. [3]

- (ii) By sketching a suitable pair of graphs, show that this equation has exactly one root in the interval $0 < x < \frac{1}{2}\pi$. [2]

- (iii) Use the iterative formula

$$x_{n+1} = \sin(2x_n),$$

with initial value $x_1 = 1$, to find α correct to 2 decimal places, showing the result of each iteration. [3]

6 The complex numbers $1 + 3i$ and $4 + 2i$ are denoted by u and v respectively.

(i) Find, in the form $x + iy$, where x and y are real, the complex numbers $u - v$ and $\frac{u}{v}$. [3]

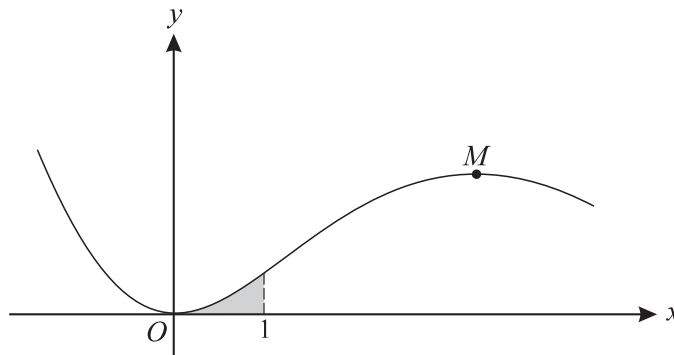
(ii) State the argument of $\frac{u}{v}$. [1]

In an Argand diagram, with origin O , the points A , B and C represent the numbers u , v and $u - v$ respectively.

(iii) State fully the geometrical relationship between OC and BA . [2]

(iv) Prove that angle $AOB = \frac{1}{4}\pi$ radians. [2]

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The diagram shows the curve $y = x^2 e^{-\frac{1}{2}x}$.

(i) Find the x -coordinate of M , the maximum point of the curve. [4]

(ii) Find the area of the shaded region enclosed by the curve, the x -axis and the line $x = 1$, giving your answer in terms of e . [5]

8 An appropriate form for expressing $\frac{3x}{(x+1)(x-2)}$ in partial fractions is

$$\frac{A}{x+1} + \frac{B}{x-2},$$

where A and B are constants.

(a) Without evaluating any constants, state appropriate forms for expressing the following in partial fractions:

(i) $\frac{4x}{(x+4)(x^2+3)}$, [1]

(ii) $\frac{2x+1}{(x-2)(x+2)^2}$. [2]

(b) Show that $\int_3^4 \frac{3x}{(x+1)(x-2)} dx = \ln 5$. [6]

- 9 The lines l and m have vector equations

$$\mathbf{r} = 2\mathbf{i} - \mathbf{j} + 4\mathbf{k} + s(\mathbf{i} + \mathbf{j} - \mathbf{k}) \quad \text{and} \quad \mathbf{r} = -2\mathbf{i} + 2\mathbf{j} + \mathbf{k} + t(-2\mathbf{i} + \mathbf{j} + \mathbf{k})$$

respectively.

- (i) Show that l and m do not intersect. [4]

The point P lies on l and the point Q has position vector $2\mathbf{i} - \mathbf{k}$.

- (ii) Given that the line PQ is perpendicular to l , find the position vector of P . [4]

- (iii) Verify that Q lies on m and that PQ is perpendicular to m . [2]

- 10 A rectangular reservoir has a horizontal base of area 1000 m^2 . At time $t = 0$, it is empty and water begins to flow into it at a constant rate of $30 \text{ m}^3 \text{ s}^{-1}$. At the same time, water begins to flow out at a rate proportional to $\sqrt{h}$, where $h \text{ m}$ is the depth of the water at time $t \text{ s}$. When $h = 1$, $\frac{dh}{dt} = 0.02$.

- (i) Show that h satisfies the differential equation

$$\frac{dh}{dt} = 0.01(3 - \sqrt{h}). \quad [3]$$

It is given that, after making the substitution $x = 3 - \sqrt{h}$, the equation in part (i) becomes

$$(x - 3)\frac{dx}{dt} = 0.005x.$$

- (ii) Using the fact that $x = 3$ when $t = 0$, solve this differential equation, obtaining an expression for t in terms of x . [5]

- (iii) Find the time at which the depth of water reaches 4 m . [2]